

Nicole Tan

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EDUCATION

North Carolina State University | Raleigh, NC

B.S. in Computer Science

May 2027

Minors: Graphic Communications, Business Administration

Relevant Coursework: Discrete Mathematics, Introduction to Java, C and Software Tools, Data Structures and Algorithms, Software Development Fundamentals, Statistics and Probability, Website Development, Operating Systems, Software Engineering

Leadership & Involvement: Vice President of Women in Computer Science (WiCS); Computer Science Ambassador; Honors Ambassador

PROJECTS

Zellit | Software Engineer & Co-founder

Present

- Built Zellit MVP as a **mobile-first marketplace app** with **React Native**, **Expo**, and **Expo Router**.
- Developed backend with **Node.js** and **Express.js**, integrating **Firebase Authentication** for .edu logins, **Cloud Firestore** for listings, and **Firestore Storage** for images.
- Assisted in designing **UI/UX prototypes** in **Figma** to support a consistent app experience and thoughtful user experience.

FAFSA Friends | PearlHacks

Feb 2025

- Built a mobile app to simplify the FAFSA process using **React Native**, **TypeScript**, and **Streamlit**.
- Integrated **Google Cloud APIs**, **Firebase**, and a **Gemini** chatbot for data handling, secure authentication, and finance support.
- Developed backend logic in JavaScript and collaborated with a team to complete the project in under 24 hours.

EcoQuest | Hack NC State

Feb 2025

- EcoQuest is designed to help users plan the most efficient routes by combining walking, biking, and bus travel to promote sustainable and accessible transportation.
- Created an iOS app in **Swift** using **MapKit** and **CoreLocation** for real-time location tracking and route visualization.
- Worked with 3 other team members to create a user-friendly interface during a 24-hour hackathon, using **Figma** for prototyping.

Project Secure | HackNC

Nov 2024

- Built a discreet, customizable safety app that lets users quietly get help by appearing as a harmless food ordering service app.
- Designed the **UI** in **Figma** to look like a food ordering app with a toggle button to reveal hidden help options.
- Worked in a team of four to develop and test a **Java command-line** simulation for core features like silent emergency signaling

PROFESSIONAL EXPERIENCE

Computer Science Instructor | The Engineering Place | Raleigh, NC

June 2025 - Aug 2025

- Taught hands-on lessons in **Python**, **AI**, and **cybersecurity** to over 200 middle and high school students across multiple camps.
- Designed and led 50+ interactive activities covering **network security**, **machine learning**, and **algorithmic problem-solving**.
- Created a collaborative and inclusive learning environment, with 90% of students reporting increased confidence in computer science topics in post-camp surveys.

Undergraduate Research Assistant | NC State University | Raleigh, NC

May 2025 - Present

- Developed and tested 20+ **Bash scripts** using **sed**, **grep**, and **awk** to automate C code transformations for **compiler fuzzing**.
- Applied regular expressions to restructure code patterns, improving test coverage and uncovering edge-case compiler bugs.
- Gained experience in low-level code analysis, compiler internals, and automated testing workflows to identify edge-case failures.

Graphic Designer | Agromeck | Raleigh, NC

Sept 2023 - Present

- Helped produce the NC State yearbook that generated over \$2,000 in sales through campus and online distribution.
- Designed visually polished yearbook spreads using **Illustrator** and **Photoshop** for a 1,000+ page award-winning publication.
- Contributed to a spread that earned an **Honorable Mention** in a national collegiate media competition.

Engineering Intern | Bosch Rexroth | Charlotte, NC

Jun 2022 - Jul 2022

- Collaborated with a cross-functional team to design and **prototype workflow solutions** using iterative problem-solving methods.
- Created **technical documentation** for onboarding, now used as a training resource for future interns.
- Developed an organizational system adopted by 100+ employees, improving **efficiency** and **scalability** of daily operations.

TECHNICAL SKILLS

- Languages:** Java, Python, C, JavaScript, SQL, R, HTML/CSS, Bash scripting, Kotlin
- Frameworks/Tools:** React, Firebase, JUnit, Jenkins, Docker, GitHub, Figma, Android Studio, JavaScript, TypeScript
- Systems:** Linux, Windows